

Casey Midgley

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EDUCATION

Worcester Polytechnic Institute

B.S./M.S. expected May 2027

B.S., Robotics Engineering and Computer Science; M.S., Robotics Engineering

Worcester, MA

- Unified Robotics, Soft Robotics, Embedded Computing, Sensors Circuits & Systems, Algorithms, Software Engineering

EXPERIENCE

Embedded Systems Engineering Intern

May 2025 — Aug 2025

Intellisplash LLC

Northborough, MA

- Developed C++ firmware for STM32-based I2C automation controller, implementing peripheral communication, device control logic, and integration support for aquatic automation hardware
- Diagnosed 2 board-level hardware faults during bring-up using firmware tests and electrical debugging, validated fixes for integration into the next PCB revision

President & Lab Manager

Jan 2023 — Jan 2026

WPI CollabLab

Worcester, MA

- Managed operations for 250+ member student-run makerspace, coordinating equipment access, workshops, member support, and lab procedures
- Negotiated relocation and renovation requirements with university administration during transition to new lab space
- Supported member projects across electronics, CAD, 3D printing, and mechanical fabrication through training, troubleshooting, and design review

PROJECTS

Formula SAE Autonomous Conversion - Goat Fast Racing

- Architected a custom Redis streams communication framework to replace ROS/ROS2, enabling modular real-time sensor, perception, localization, control, and logging pipelines for a Formula SAE autonomous EV platform.
- Led FSDS simulation infrastructure for driverless-stack development under limited vehicle access, supporting full-stack validation on Jetson AGX Orin hardware.
- Built EKF localization calibration workflow using Redis data replay, reducing tuning cycles from several hours to ~3 minutes and improving pose-estimation fitness score from 13 to 3.25.
- Designed STM32 watchdog-board infrastructure to monitor stale sensor data, localization faults, and system-health failures for controlled shutdown behavior.
- Integrated perception and localization components across YOLO/OpenCV, LiDAR, GNSS/IMU, and EKF pipelines, establishing a reusable autonomy foundation for future vehicle development.

Chip8Emu & ChipBoy8 - CHIP-8 Emulator and STM32G4 Handheld Console

- Built modular CHIP-8 emulator in C++ with SDL2-based graphics/input, memory visualization, and debugging tools
- Improved emulator throughput from 1 Hz to 200 Hz by profiling SDL2/debug-rendering bottlenecks and optimizing code paths for decoupled updates on memory and UI regions.
- Architected the emulator core for portability across desktop, browser, and embedded handheld targets, including the planned STM32G4-based ChipBoy8 console

Custom Combat Robot - Modular Plastic Antweight Robot

- Designed and built modular combat robot for competition, optimizing mechanical and electrical architecture for rapid repair after match damage.
- Reduced component replacement time by ~90%, enabling any major component to be replaced in under 1 minute.
- Validated design through competition use under impact, vibration, and time-constrained repair conditions.

SKILLS

- **Languages:** C/C++, Python, TypeScript/JavaScript, Java, MATLAB
- **Embedded / Robotics:** STM32, ESP32, I2C, SPI, UART, PWM, GPIO, PlatformIO, ROS, OpenCV, EKF
- **Hardware / Tools:** Git, GDB, Redis, KiCad, Fusion 360, Onshape, FDM/SLA 3D printing, DFM, CAM